
Organizing ROS Code

ROS Packages

- ROS packages group multiple nodes, libraries, and other resource together for specific purpose or category of functionality
- Packages are the buildable and redistribuatl units ROS code

Build Tools

- Code needs to be compiled into a form that computers can understand
 - This is accomplished with a compiler (g++, MSVC, etc.)
- Compilers require special flags and work on one library/program at a time
- Build systems help run compilers on full projects, but generally use machine-specific values (like full absolute file paths to library files)
 - make, ninja, etc.
- Tools like Cmake can be used to generate build tool files from higher level dependency relationships
 - ROS uses a special extension of Cmake called ament_cmake that adds some extra functionality to make it easier to work with ROS packages
- colcon provides an interface to build, test, and inspect many packages at once on the workspace level

